

- A remote repository exists outside of your laptop, somewhere on a remote server. In our case, it will be on GitHub.
- **Git objects:** There are different kinds of objects that Git uses such as:
 - **Tree**—This contains one or more ‘trees’ and blobs. A tree is a sort of a file system directory and it can point to other Git trees. Think of it as a directory having other sub-directories or ‘blobs’.
 - **Blobs**—These are just normal plain files such as text files, source code files, or even pictures; something like the file contents or the *inode* information. Basically, a blob can be thought of as something that’s not a directory.
 - **Commit**—This is generated the moment you run the *git commit* command. It has the meta data as well as a pointer to the root project directory, so that the snapshots can be recreated whenever needed.
- **Git index:** This is the staging area of Git. When you run the *git add* command, you add files to this staging area. When you run the *git commit* command, these files are committed to your local Git database. You then push the files into the remote repository.



Note: The index is NOT the repository, neither is it the working tree/directory.

- **Git commit:** This is a point in time snapshot of your working directory. You run the *git commit* command and the files are committed to the local Git database as stated earlier. This command basically checksums all the files and directories in the location where it is run and creates a commit object.
- **Branch:** It is a pointer to a commit. The default branch name in Git is ‘master’. The master branch appears the first time we do a commit. If we do another commit, the master then points to this new commit. There is other terminology, but this is enough for starters.

Opening a GitHub account

So let’s open a GitHub account first. Head to www.github.com and sign up. It’s free.



Note: Please note down the e-mail address and the password you use to create this account. We will need it in a while when we configure Git on the laptop or desktop.

Once registration is complete, you can sign in. On the top right side of the page, you will see the name you chose and there will be a ‘+’ sign next to it. Click on it and then on the new repository option. Take a look at the screenshot in Figure 1.

On the next page, choose any name for your repository. Mine’s called *mydotfiles*. Provide a brief description about your repository. Please do not select the ‘Initialise this repository with a README’ option because we will be

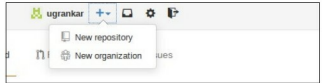


Figure 1: Create repo



Figure 2: Git repo

creating a README ourselves.

Click on the ‘Create repository’ button and when the next page loads, make a note of the URL shown.

Figure 3 shows how it looks in my case. This is a screenshot of the top half of the page that loads up when you hit the ‘Create repository’ button showing the Git remote repository name.

Important note: Please remember, whatever repository name you choose, you’ll either need to have a directory with exactly the same name or, you’ll need to create a new directory with the same name. Git is basically used to sync your directories and files located inside a directory on your laptop with the one having the same name on GitHub.

Installing and configuring Git for first time use

Install Git by running:

```
yum install git
```

Once you have it installed, run the *git --version* command to see which version we are on. This is how it looks on my laptop:

```
[pmu@t430 ~]$ git --version
git version 1.8.3.1
[pmu@t430 ~]$
```

Now, let’s configure Git for first time use. We will set up the name and e-mail address on our local laptop or desktop. This is the step we have to run the very first time we set up Git on a laptop or desktop.



Figure 3: Git remote address